KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2021
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 3 hours FULL MARKS: 210

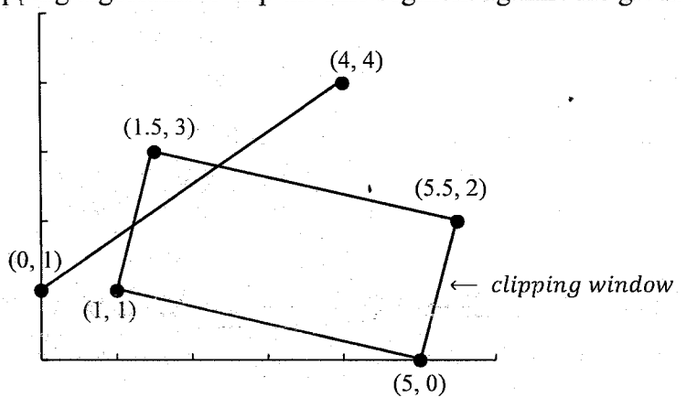
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Briefly explain the different functionalities of 3D computer graphics pipeline. (10)
   b) Show that two successive reflections about any two lines in the xy plane that go through the origin is equivalent to a rotation about the origin. (12)
   c) Use quaternion mathematics to derive a 3D rotation matrix about a unit vector $\boldsymbol{u}$. (13)

2. a) Derive a perspective projection matrix if the center of projection is at (a, b, c), the projection plane normal $N = n_1 i + n_2 j + n_3 k$ and reference point $R_0(x_0, y_0, z_0)$. (13)
   b) Determine the oblique projection of a VCS point $P(2, 2, 5)$ onto a view plane with $N = k$ and $R_0(1, 0, 1)$. Use azimuth = 30° and elevation = 45°. (10)
   c) Consider the following figure showing a rectangular clipping window and a line segment. Apply Cyrus-Beck line clipping algorithm to clip the line segment against the given window. Find intersection points. (12)

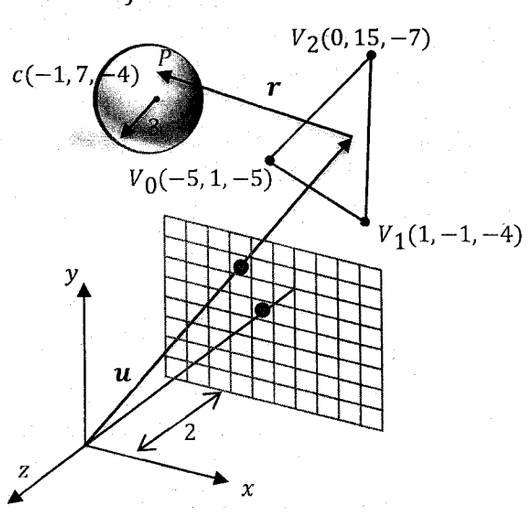


3. a) There are two monochromatic light sources L^0 and L^1 in a scene. The light intensities and corresponding material's reflectivity at vertex V are given in the following table. (12)

	Ambient	Diffuse	Specular
L^0	0.1	0.9	0.9
L^1	0.2	0.7	0.5
V	0.2	0.6	1.0

Furthermore, the shininess at $V = 3.0$, vertex normal at V is $[0 \quad 1 \quad 0]^T$, emissive light at V is 0.05. The position vectors of L^0, L^1, V and eye are, respectively, $[0 \quad 5 \quad 5]^T$, $[5 \quad 5 \quad 5]^T$, $[0 \quad 0 \quad 5]^T$ and $[0 \quad 0 \quad 0]^T$. Use Blinn-Phong illumination model to approximate the total intensity at V.

   b) Consider, the information given in the following figure. Determine whether the reflected ray $\boldsymbol{r}$ intersects the sphere at P. If so, find the P. Here, it is given that the approximated triangular surface normal $N = -1.04i + 1.77j + 9.79k$. (12)



   c) Define and explain the basic radiosity equation. State a method to solve the equation. (11)

4. a) Find Camera Coordinate System (CCS) value for a world point $P(1, 2, 3)$ if the camera is at $(3, 3, 7)$ focusing towards $(1, 1, 1)$ with up vector $U = i + j$. (12)
   b) Write down the 4×4 rotation matrix M that transforms the orthonormal 3D vectors $\boldsymbol{u}(x_u, y_u, z_u)$, $\boldsymbol{v}(x_v, y_v, z_v)$, and $\boldsymbol{w}(x_w, y_w, z_w)$ to orthonormal 3D vectors $\boldsymbol{a}(x_a, y_a, z_a)$, $\boldsymbol{b}(x_b, y_b, z_b)$, and $\boldsymbol{c}(x_c, y_c, z_c)$ such that $M\boldsymbol{u} = \boldsymbol{a}$, $M\boldsymbol{v} = \boldsymbol{b}$, and $M\boldsymbol{w} = \boldsymbol{c}$. (16)
   c) Define z-fighting problem. Mention some strategies to overcome the issue. (07)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define horizontal and vertical retrace. (04)
   b) If a circle's origin is centered at point (5, 5) and radius is equivalent to 5, then rasterize the circle using midpoint circle drawing algorithm. (14)
   c) If $(x_0, y_0) = (5, 15)$ and $(x_{end}, y_{end}) = (11, 9)$, use Bresenham's algorithm to compute pixels for line segment. (09)
   d) What are the major side effects of scan conversion? Explain. (08)

6. a) Write down some limitations of z-Buffer algorithm. Explain how A-Buffer method overcomes the drawback of z-Buffer. (08)
   b) Explain the basic idea of Painter's algorithm. How does the algorithm deal with the situation when the polygons are such that their x, y and z-extents overlap? (10)
   c) Consider two triangles, ΔABC defined by the points $A(1, 7, 5)$, $B(1, 1, 5)$, $C(6, 1, 5)$ and ΔDEF defined by the points $D(6, 5, 2)$, $E(2, 0, 3)$, $F(7, 0, 7)$. Apply z-Buffer algorithm to the triangle surface using a 8×8 depth buffer. Show z-Buffer contents for all the pixels. Also show the frame buffer content if the surface colors of ΔABC and ΔDEF are RED and GREEN respectively. (17)

7. a) Equation of two curves Q and R are given as follows: (09)

$$Q(t) = (1 + t^2)P_1 + (5t - 5t^2)P_2 + t^2P_3$$
$$R(t) = (1 + t^2)P_3 + (5t - 5t^2)P_4 + t^2P_5$$

   For both the curves $0 \leq t \leq 1$. Are both the curves C^0 continuous at $Q(1)$ and $R(0)$? Are both the curves C^1 and G^1 continuous at $Q(0)$ and $R(0)$? Why or why not? Explain your answer.
   b) Show that a cubic Bézier curve can be presented as following matrix formulation; symbols have their usual meanings. (14)

$$P(t) = [t^3 \quad t^2 \quad t \quad 1] \begin{bmatrix} -1 & 3 & -3 & 1 \\ 3 & -6 & 3 & 0 \\ -3 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} P_1 \\ P_{C_1} \\ P_{C_2} \\ P_2 \end{bmatrix}$$

   Also draw a rough sketch of the curve if $P_1(0, 0)$, $P_{C_1}(0, 3)$, $P_{C_2}(2, 0)$ and $P_2(4, 0)$.
   c) An open uniform B-spline curve $Q(u)$ of order $K = 3$ is constructed with the control points $P_0(0, 0, 0)$, $P_1(1, 1, 0)$, $P_2(2, 0, 0)$, $P_3(3, 1, 0)$ and $P_4(4, 0, 0)$. Determine the knot vector and parametric range for each segment of the curve. Also determine the blending functions for the last segment of the curve. (12)

8. a) How does the texture magnification effect and texture minification effect occur? How to determine the appropriate mipmap level in mipmapping? (05)
   b) How does a CRT monitor display color pictures? Briefly explain. (07)
   c) Define fractal. Generate a fractal image using the following language. (12)

$$AA[A[B]AA(B)]AAAA(A[B]AA(B))$$

   where Alphabet: $\{A, B, [,], (,)\}$,
   Rule: $A \Rightarrow AA$, $B \Rightarrow A[B]AA(B)$, $[] \Rightarrow left\ brance$, $() \Rightarrow right\ brance$.
   d) Consider the scan line Fill algorithm. Determine the content of active edge table to fill the polygon. What are the global edge table for the polygon? (11)

